MIDTERM LABORATORY ACTIVITY

Portfolio CRUD with Validation

Laravel + Inertia.js + React

Activity detail	Value
Student	Mark-dj Villar
Course	IT ELEC01 ? Advanced Web Development
Section	[Fill in your class section]
Instructor	Giobet T. Aban
Date	November 1, 2026

Review the reflection drafts, add the folder-structure screenshot, and insert the final GitHub and video links before submission.

1. Title Page

Activity: Midterm Laboratory Activity ? Portfolio CRUD with Validation. Student: Mark-dj Villar. Course: IT ELEC01 ? Advanced Web Development. Instructor: Giobet T. Aban. Class section and submission links must be filled in by the student.

2. Section Chosen and Justification

Chosen section: Projects. Projects are work samples that change over time, so storing them in a database makes them editable without changing source code. Each record can include a category, description, tools, links, an image, and a completion date; the public portfolio reads the current records from the database.

Fields and data types

Field	Database type	Purpose
title	string(60), unique	Project name; uniqueness prevents duplicate titles.
category_id	foreign key / integer	Links each project to a reusable category.
description	text	Stores a short project overview.
tools	string(80)	Lists technologies used in the project.
project_url	nullable string	Optional link to the live project.
repository_url	nullable string	Optional link to the source repository.
image	string	Stores the public-storage path of the uploaded image.
date_completed	date	Records when the project was completed.

Validation rules include required values, length limits, URL/date formats, unique title checks that ignore the edited record, image type/size checks, and case-insensitive duplicate-category checks. The exact rules are implemented in ProjectController.

3. Screenshots of Required Pages

The following labeled screenshots were captured from the running local application. They show the database-backed Projects section and the current UI.

Figure 1. Public portfolio page showing the dynamic Projects section.

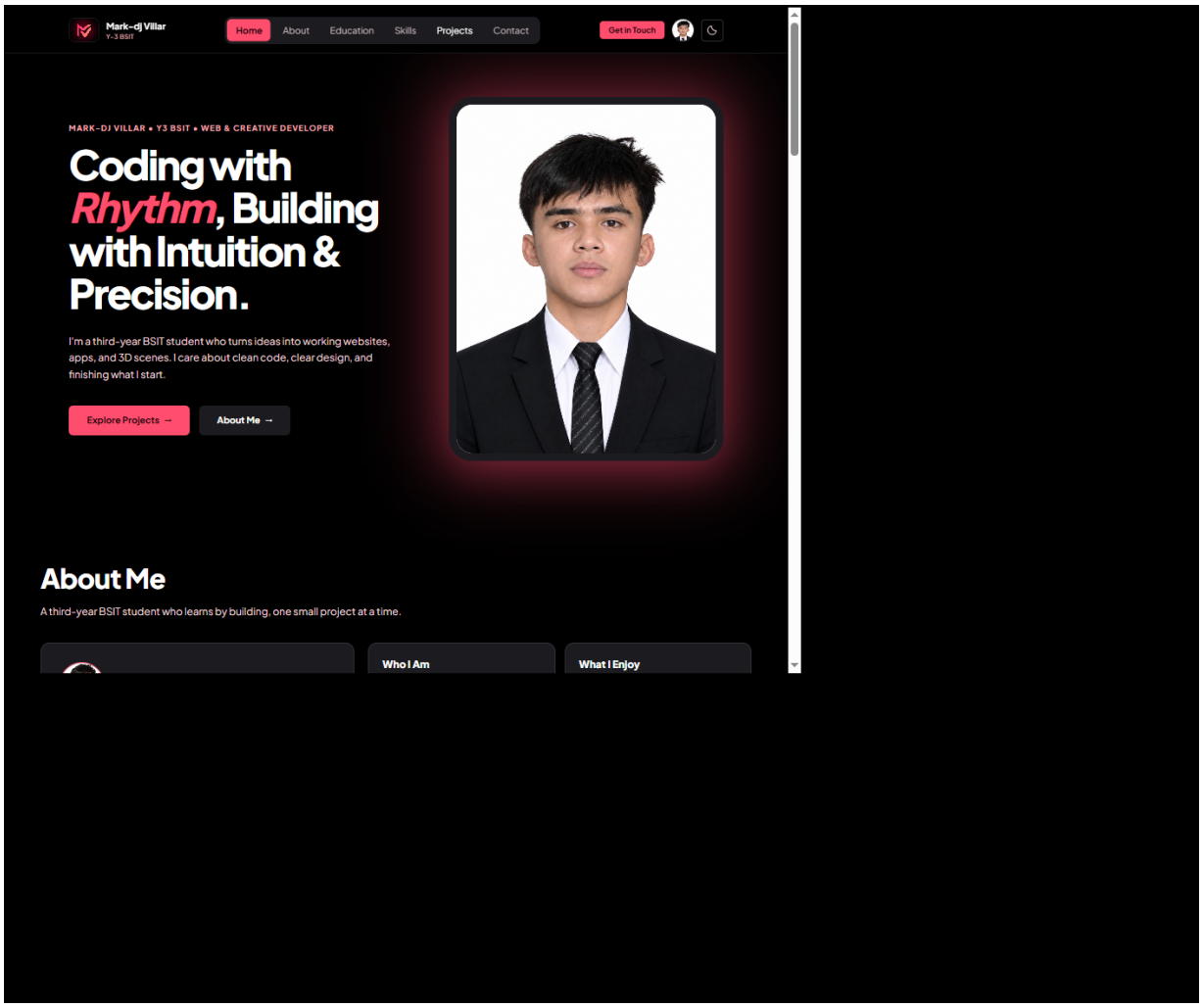


Figure 2. Projects index page with the existing portfolio record.

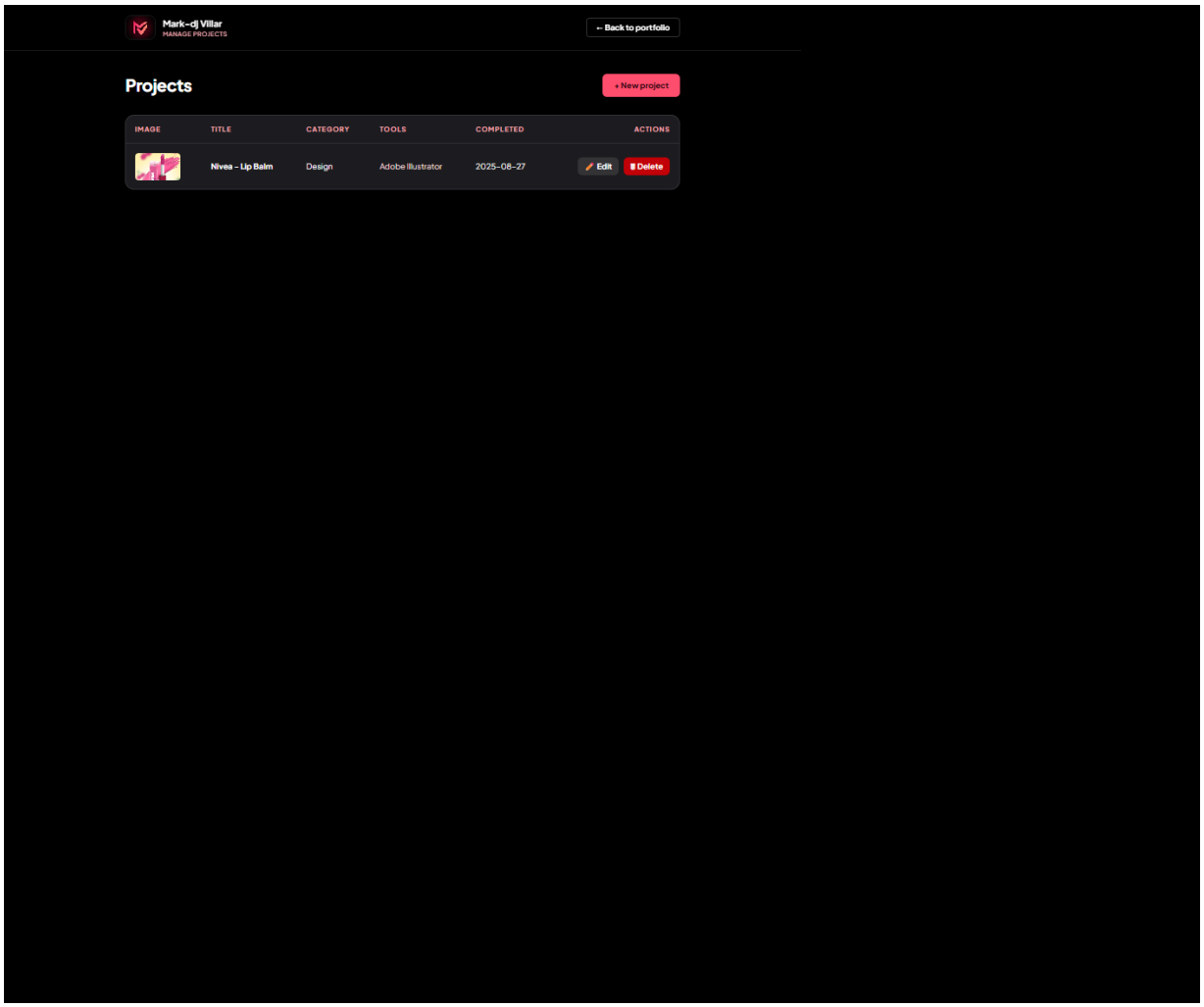


Figure 3. Create Project page.

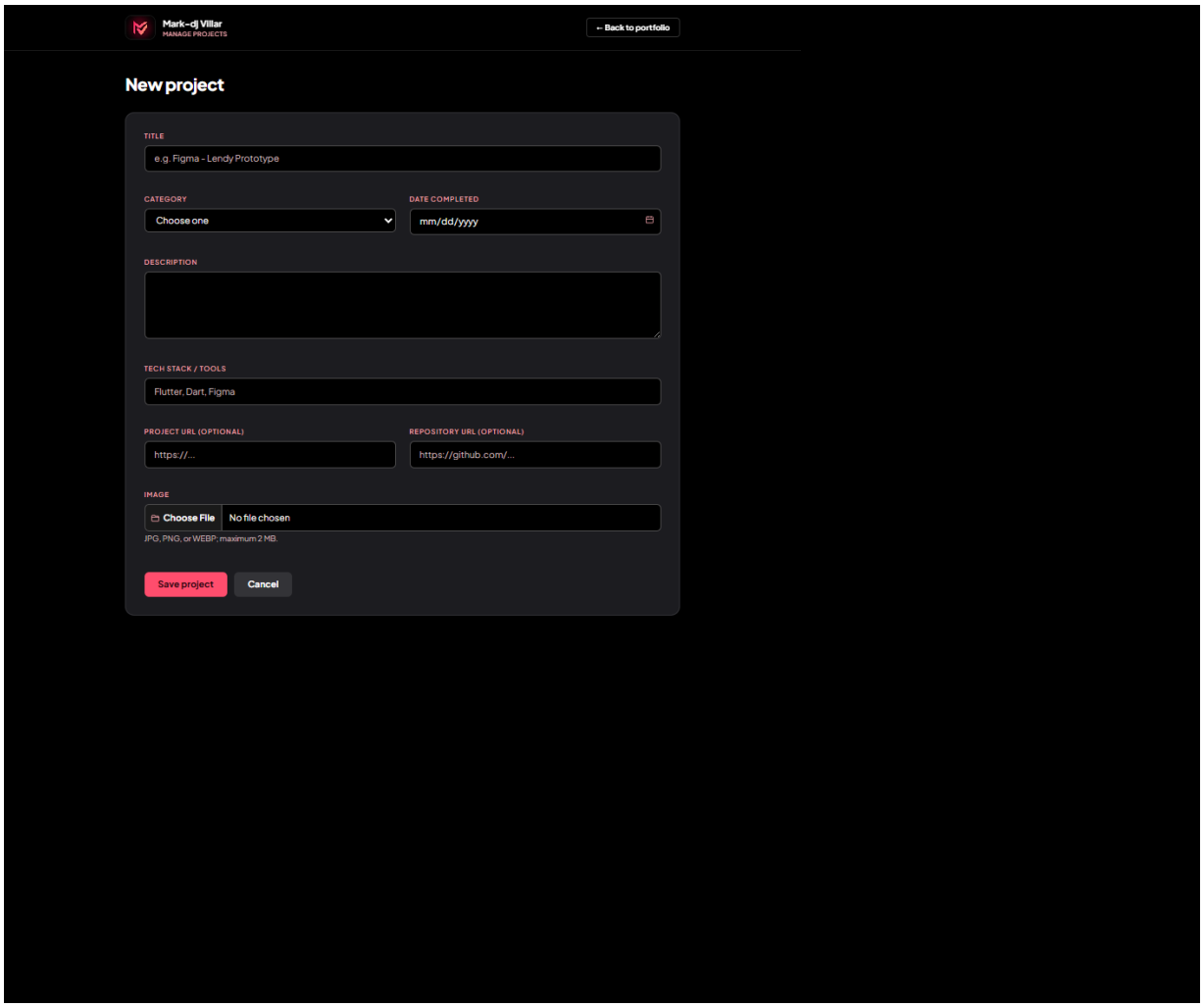


Figure 4. Edit Project page.

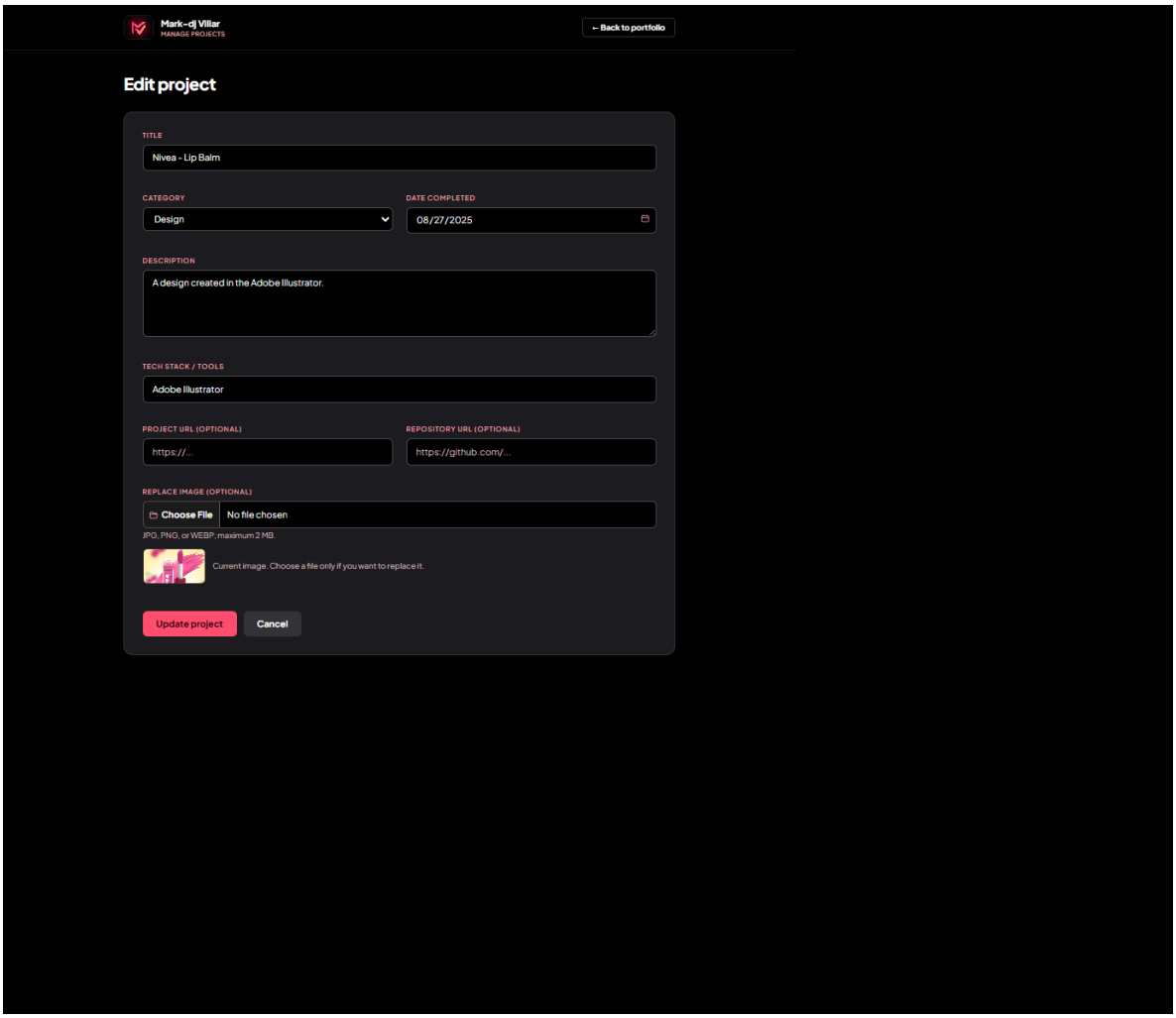


Figure 5. Custom delete confirmation dialog.

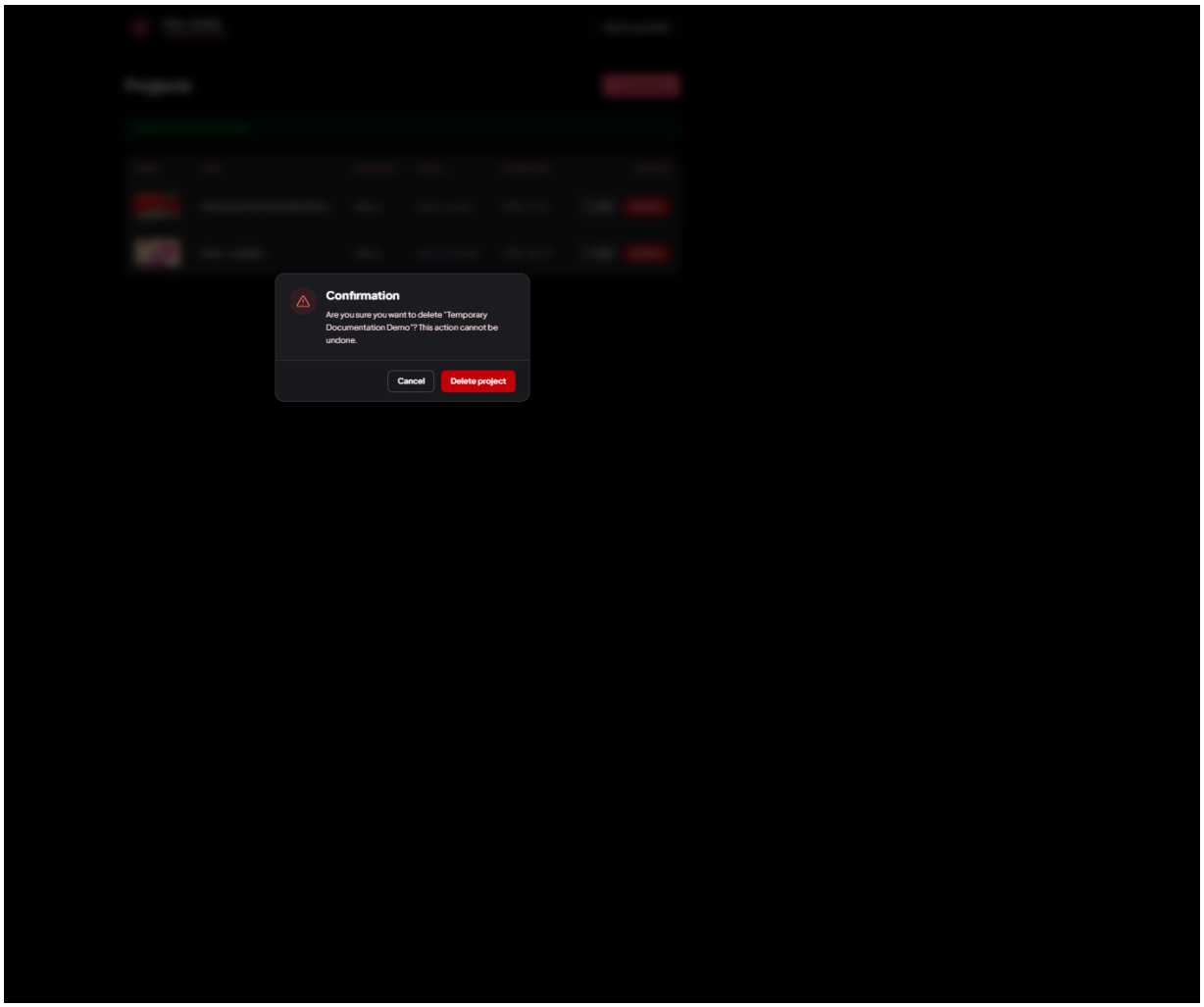


Figure 6. Success message after creating a project.

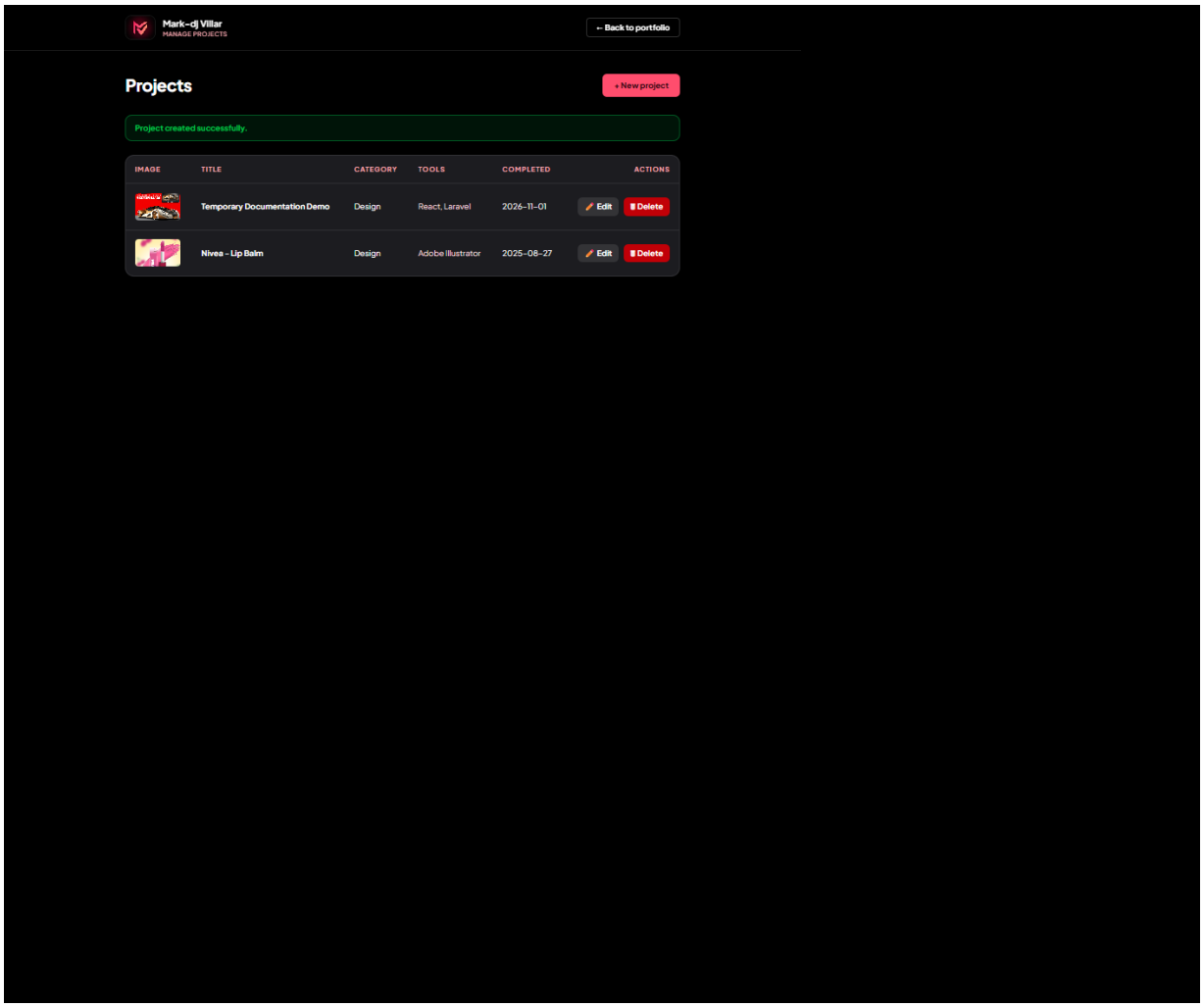
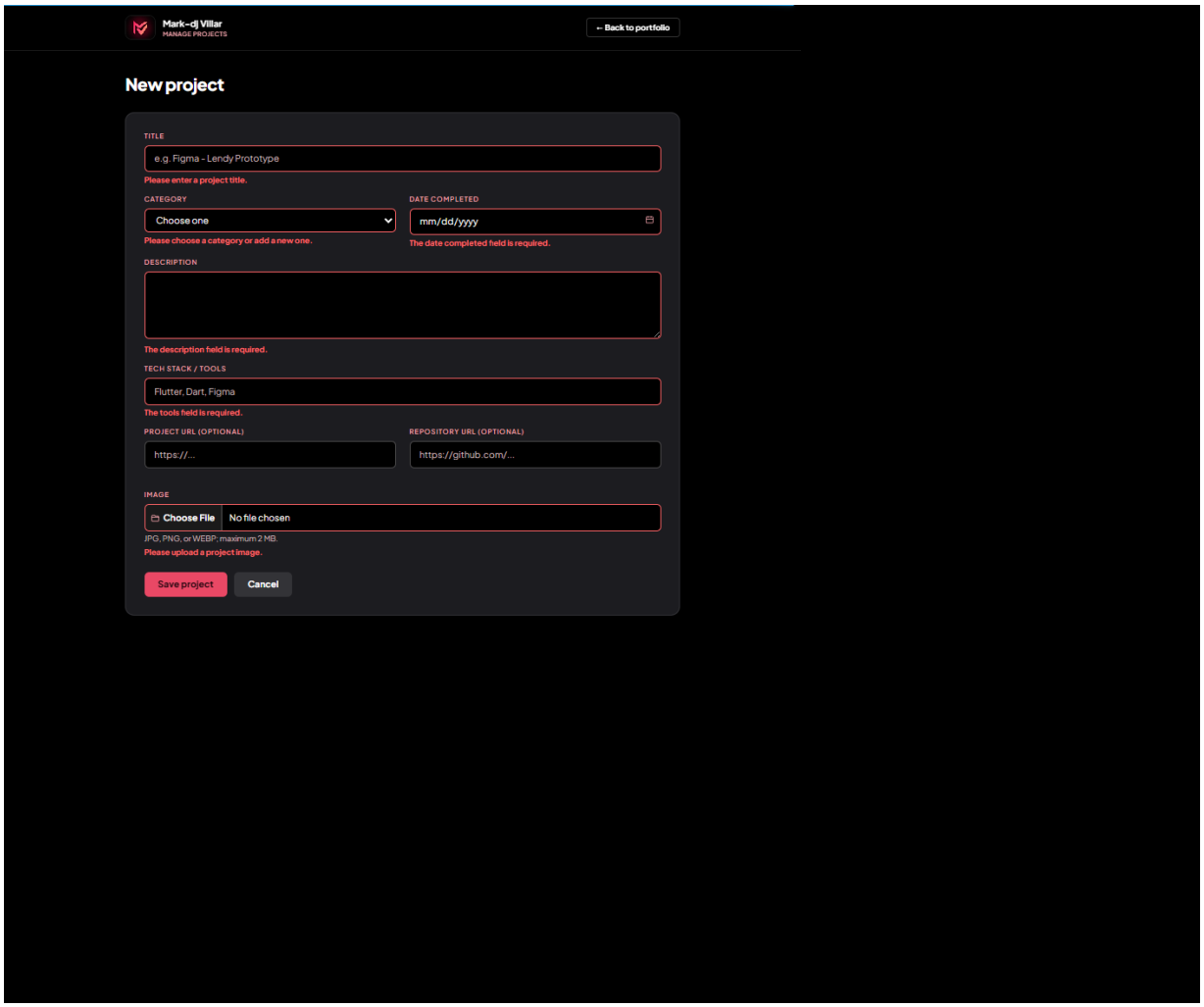


Figure 7. Validation errors shown next to the related fields.



Additional screenshot to add: the VS Code Explorer or another file browser showing the project folder structure. This screenshot is not included in this generated document.

4. Folder Structure

The relevant files are organized as follows. Capture an Explorer screenshot of this structure and add it before submission.

```
portfolio CRUD/
|-- app/
|   |-- Http/Controllers/
|   |   |-- PortfolioController.php
|   |   `-- ProjectController.php
|   |-- Http/Middleware/HandleInertiaRequests.php
|   `-- Models/Category.php, Project.php
|-- database/migrations/
|   |-- *_create_categories_table.php
|   `-- *_create_projects_table.php
|-- resources/
|   |-- css/app.css
|   |-- data/portfolio.php
|   |-- js/app.jsx
|   |-- js/Components/ (reusable React components)
|   |-- js/Pages/Portfolio.jsx
|   |-- js/Pages/Projects/ (Index, Create, Edit)
|   `-- views/app.blade.php
|-- routes/web.php
`-- tests/Feature/ (CRUD and validation tests)
```

SQLite database: database/database.sqlite. Uploaded project images: storage/app/public/projects, exposed through public/storage.

5. Complete Source Code

The following appendix includes the complete source of the migration, models, controllers, routes, Inertia root, React pages, reusable components, styling, and feature tests. Read each short explanation first, then follow the full source listing.

database/migrations/2026_11_01_000001_create_categories_table.php

Creates the category table and unique category names.

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('categories', function (Blueprint $table) {
            $table->id();
            $table->string('name', 30)->unique(); // created by the user, never hard-coded
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('categories');
    }
};
```

database/migrations/2026_11_01_000002_create_projects_table.php

Defines project fields and the category foreign key.

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    public function up(): void
    {
        Schema::create('projects', function (Blueprint $table) {
            $table->id();
            $table->foreignId('category_id')->constrained()->cascadeOnDelete();
            $table->string('title', 60)->unique();
            $table->text('description');
            $table->string('tools', 80);
            $table->string('project_url')->nullable();
            $table->string('repository_url')->nullable();
            $table->string('image');
            $table->date('date_completed');
            $table->timestamps();
        });
    }

    public function down(): void
    {
        Schema::dropIfExists('projects');
    }
};
```

app/Models/Category.php

Defines category mass assignment and the one-to-many project relation.

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\HasMany;

class Category extends Model
{
    protected $fillable = ['name'];

    public function projects(): HasMany
    {
        return $this->hasMany(Project::class);
    }
}
```

app/Models/Project.php

Defines fillable project fields, date/image accessors, and the category relation.

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Casts\Attribute;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\BelongsTo;

class Project extends Model
{
    // Only these columns can be mass-assigned (Project::create / ->update).
    protected $fillable = [
        'category_id',
        'title',
        'description',
        'tools',
        'project_url',
        'repository_url',
        'image',
        'date_completed',
    ];

    // Sent to React together with the normal columns.
    protected $appends = ['image_url'];

    protected function casts(): array
    {
        return ['date_completed' => 'date:Y-m-d'];
    }

    public function category(): BelongsTo
    {
        return $this->belongsTo(Category::class);
    }

    // /storage/projects/xxxx.png (needs: php artisan storage:link)
    protected function imageUrl(): Attribute
    {
        return Attribute::get(fn () => '/storage/'.$this->image);
    }
}
```

app/Http/Controllers/PortfolioController.php

Loads database projects and static portfolio content for the public React page.

```php
<?php

namespace App\Http\Controllers;

use App\Models\Project;
use Inertia\Inertia;
use Inertia\Response;

class PortfolioController extends Controller
{
    public function home(): Response
    {
        return Inertia::render('Portfolio', [
            'projects' => Project::with('category')->latest()->get(),
            'portfolio' => require resource_path('data/portfolio.php'),
        ]);
    }
}
```

app/Http/Controllers/ProjectController.php

Implements resource CRUD, server validation, category creation, and safe image replacement/deletion.

```
<?php

namespace App\Http\Controllers;

use App\Models\Category;
use App\Models\Project;
use Illuminate\Http\RedirectResponse;
use Illuminate\Http\Request;
use Illuminate\Support\Arr;
use Illuminate\Support\Facades\DB;
use Illuminate\Support\Facades\Storage;
use Illuminate\Validation\Rule;
use Inertia\Inertia;
use Inertia\Response;
use RuntimeException;
use Throwable;

class ProjectController extends Controller
{
    public function index(): Response
    {
        return Inertia::render('Projects/Index', [
            'projects' => Project::with('category')->latest()->get(),
        ]);
    }

    public function create(): Response
    {
        return Inertia::render('Projects/Create', [
            'project' => null,
            'categories' => Category::orderBy('name')->get(['id', 'name']),
        ]);
    }

    public function store(Request $request): RedirectResponse
    {
        $this->normalizeNewCategorySelection($request);
        $data = $request->validate($this->rules(), $this->messages());
        $imagePath = $request->file('image')->store('projects', 'public');
        if ($imagePath === false) {
            throw new RuntimeException('Unable to store the project image.');
        }

        try {
            DB::transaction(function () use ($data, $imagePath): void {
                Project::create([
                    ...Arr::only($data, ['title', 'description', 'tools', 'project_url',
'repository_url', 'date_completed']),
                    'category_id' => $this->resolveCategoryId($data),
                    'image' => $imagePath,
                ]);
            });
        } catch (Throwable $exception) {
            Storage::disk('public')->delete($imagePath);
            throw $exception;
        }

        return to_route('projects.index')->with('success', 'Project created successfully.');
    }

    public function edit(Project $project): Response
    {
        return Inertia::render('Projects/Edit', [
            'project' => $project,
            'categories' => Category::orderBy('name')->get(['id', 'name']),
        ]);
    }

    public function update(Request $request, Project $project): RedirectResponse
    {
        $this->normalizeNewCategorySelection($request);
        $data = $request->validate($this->rules($project), $this->messages());
        $oldImagePath = $project->image;
        $newImagePath = $request->hasFile('image')
            ? $request->file('image')->store('projects', 'public')
            : null;
        if ($newImagePath === false) {
            throw new RuntimeException('Unable to store the replacement project image.');
        }

        try {
            DB::transaction(function () use ($data, $newImagePath, $project): void {
                $payload = [
                    ...Arr::only($data, ['title', 'description', 'tools', 'project_url',
'repository_url', 'date_completed']),
                    'category_id' => $this->resolveCategoryId($data),
                ];

                if ($newImagePath !== null) {
```

```
                $payload['image'] = $newImagePath;
            }

            $project->update($payload);
        });
    } catch (Throwable $exception) {
        if ($newImagePath !== null) {
            Storage::disk('public')->delete($newImagePath);
        }

        throw $exception;
    }

    if ($newImagePath !== null) {
        Storage::disk('public')->delete($oldImagePath);
    }

    return to_route('projects.index')->with('success', 'Project updated successfully.');
}

public function destroy(Project $project): RedirectResponse
{
    $imagePath = $project->image;
    $project->delete();
    Storage::disk('public')->delete($imagePath);

    return back()->with('success', 'Project deleted successfully.');
}

private function rules(?Project $project = null): array
{
    return [
        'title' => ['required', 'string', 'max:60', Rule::unique('projects',
'title')->ignore($project?->id)],
        'category_id' => ['nullable', 'required_without:new_category', 'exists:categories,id'],
        'new_category' => [
            'nullable',
            'required_without:category_id',
            'string',
            'max:30',
            function (string $attribute, mixed $value, \Closure $fail): void {
                if ($value && Category::whereRaw('LOWER(name) = ?',
[mb_strtolower(trim($value))])->exists()) {
                    $fail('That category already exists.');
                }
            },
        ],
        'description' => ['required', 'string', 'min:20', 'max:300'],
        'tools' => ['required', 'string', 'max:80'],
        'project_url' => ['nullable', 'url', 'max:255'],
        'repository_url' => ['nullable', 'url', 'max:255'],
        'date_completed' => ['required', 'date', 'before_or_equal:today'],
        'image' => [$project ? 'nullable' : 'required', 'image', 'mimes:jpg,jpeg,png,webp', 'max:
2048'],
    ];
}

private function messages(): array
{
    return [
        'title.required' => 'Please enter a project title.',
        'title.unique' => 'You already have a project with this title.',
        'category_id.required_without' => 'Please choose a category or add a new one.',
        'new_category.required_without' => 'Enter the name of the new category.',
        'description.min' => 'The description must be at least :min characters.',
        'date_completed.before_or_equal' => 'The date completed cannot be in the future.',
        'image.required' => 'Please upload a project image.',
        'image.image' => 'The file must be an image.',
        'image.mimes' => 'The image must be a JPG, PNG or WEBP file.',
        'image.max' => 'The image must not be larger than 2 MB.',
    ];
}

private function resolveCategoryId(array $data): int
{
    if (! empty($data['new_category'])) {
        return Category::create(['name' => trim($data['new_category'])])->id;
    }

    return (int) $data['category_id'];
}

private function normalizeNewCategorySelection(Request $request): void
{
    if ($request->input('category_id') === '__new') {
        $request->merge(['category_id' => null]);
    }
}
}
```

app/Http/Middleware/HandleInertiaRequests.php

Shares success flash messages with every Inertia page.

```php
<?php

namespace App\Http\Middleware;

use Illuminate\Http\Request;
use Inertia\Middleware;

class HandleInertiaRequests extends Middleware
{
    protected $rootView = 'app';

    public function share(Request $request): array
    {
        return [
            ...parent::share($request),
            'flash' => [
                'success' => fn () => $request->session()->get('success'),
            ],
        ];
    }
}
```

bootstrap/app.php

Registers the Inertia middleware in Laravel web requests.

```
<?php

use Illuminate\Foundation\Application;
use Illuminate\Foundation\Configuration\Exceptions;
use Illuminate\Foundation\Configuration\Middleware;
use App\Http\Middleware\HandleInertiaRequests;

return Application::configure(basePath: dirname(__DIR__))
    ->withRouting(
        web: __DIR__.'/../routes/web.php',
        commands: __DIR__.'/../routes/console.php',
        health: '/up',
    )
    ->withMiddleware(function (Middleware $middleware): void {
        $middleware->web(append: [
            HandleInertiaRequests::class,
        ]);
    })
    ->withExceptions(function (Exceptions $exceptions): void {
        //
    })->create();
```

routes/web.php

Maps the portfolio home and resource CRUD URLs to their controllers.

```
<?php

use App\Http\Controllers\PortfolioController;
use App\Http\Controllers\ProjectController;
use Illuminate\Support\Facades\Route;

Route::get('/', [PortfolioController::class, 'home'])->name('home');
Route::resource('projects', ProjectController::class)->except('show');
```

resources/views/app.blade.php

Provides the minimal Inertia root view and Vite assets.

```
<!doctype html>
<html lang="en" data-theme="dark">
<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <meta name="description" content="Portfolio of Mark-dj Villar, a BSIT student building websites,
apps, 3D scenes, and vector art.">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;700;
800&display=swap" rel="stylesheet">
    <script>
        try {
            const theme = localStorage.getItem('portfolio-theme');
            if (theme === 'light' || theme === 'dark') document.documentElement.dataset.theme = theme;
        } catch (error) {
            console.warn('Unable to restore the saved color theme.', error);
        }
    </script>
    @vite(['resources/css/app.css', 'resources/js/app.jsx'])
    @inertiaHead
</head>
<body class="bg-ink font-sans text-white antialiased">
    @inertia
</body>
</html>
```

resources/css/app.css

Defines Tailwind theme colors, accessibility focus states, and portfolio theme styles.

```
@import "tailwindcss";

@theme {
    --color-ink: #000000;
    --color-panel: #1c1c20;
    --color-coral: #ff4d6d;
    --color-peach: #f7a7af;
    --color-blush: #f3d9de;
    --color-mute: #c9b8bc;
    --color-deep: #2a0510;
    --color-rose: #df3154;
    --font-sans: "Plus Jakarta Sans", ui-sans-serif, system-ui, sans-serif;
}

:focus-visible {
    outline: 3px solid var(--color-peach);
    outline-offset: 3px;
}

html {
    scroll-behavior: smooth;
    scroll-padding-top: 5rem;
}

:root[data-theme="light"] {
    --color-ink: #f6f4f5;
    --color-panel: #ffffff;
    --color-coral: #df3154;
    --color-peach: #a52b45;
    --color-blush: #42363a;
    --color-mute: #6c5e63;
    --color-deep: #ffffff;
    --portrait-shadow: 0 18px 60px rgb(223 49 84 / 16%);
    --portrait-shadow-strong: 0 24px 80px rgb(223 49 84 / 32%);
    color-scheme: light;
}

:root[data-theme="light"] body {
    color: #211a1d;
}

:root[data-theme="light"] .text-white {
    color: #211a1d;
}

:root[data-theme="light"] [class*="border-white/"] {
    border-color: rgb(33 26 29 / 12%);
}

:root[data-theme="light"] [class*="bg-white/"] {
    background-color: rgb(33 26 29 / 6%);
}

:root[data-theme="light"] [class~="hover:text-white"]:hover {
    color: #211a1d;
}

.hero-portrait-frame {
    box-shadow: var(--portrait-shadow, 0 18px 60px rgb(255 77 109 / 24%));
    animation: portrait-shadow-pulse 3s ease-in-out infinite;
}

@keyframes portrait-shadow-pulse {
    0%, 100% { box-shadow: var(--portrait-shadow, 0 18px 60px rgb(255 77 109 / 24%)); }
    50% { box-shadow: var(--portrait-shadow-strong, 0 24px 80px rgb(255 77 109 / 42%)); }
}

.nav-indicator {
    position: absolute;
    top: 0;
    left: 0;
    z-index: 0;
    border-radius: 0.5rem;
    background-color: var(--color-coral);
    opacity: 0;
    pointer-events: none;
    transition: transform 350ms cubic-bezier(0.22, 1, 0.36, 1), width 350ms cubic-bezier(0.22, 1, 0.36,
1), height 350ms cubic-bezier(0.22, 1, 0.36, 1), opacity 150ms ease;
}

.nav-indicator.is-visible {
    opacity: 1;
}

.nav-link {
    position: relative;
    z-index: 1;
}
```

```
.image-file-input::file-selector-button {
    margin: -0.625rem 0.75rem -0.625rem -1rem;
    padding: 0.625rem 0.75rem 0.625rem 2rem;
    border: 0;
    border-right: 1px solid rgb(255 255 255 / 30%);
    background-color: rgb(255 255 255 / 10%);
    background-image: url("data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 24
24' fill='none' stroke='%23f7a7af' stroke-width='1.8' stroke-linecap='round'
stroke-linejoin='round'%3E%3Cpath d='M3 7a2 2 0 0 1 2-2h5l2 2h7a2 2 0 0 1 2 2v9a2 2 0 0 1-2 2H5a2 2 0 0
1-2-2z'/%3E%3Cpath d='M3 10h18'/%3E%3C/svg%3E");
    background-position: 0.625rem center;
    background-repeat: no-repeat;
    background-size: 1rem;
    color: white;
    font: inherit;
    font-weight: 700;
    cursor: pointer;
}

.image-file-input::file-selector-button:hover {
    background-color: rgb(255 255 255 / 20%);
}

.date-completed-input::-webkit-calendar-picker-indicator {
    position: absolute;
    inset: 0;
    width: auto;
    height: auto;
    opacity: 0;
    cursor: pointer;
}

.theme-toggle-icon {
    transition: transform 200ms ease, opacity 200ms ease;
}

.theme-icon-sun {
    display: none;
}

:root[data-theme="light"] .theme-icon-sun {
    display: block;
}

:root[data-theme="light"] .theme-icon-moon {
    display: none;
}

@media (prefers-reduced-motion: reduce) {
    html { scroll-behavior: auto; }
    body, body * { transition: none !important; }
    .nav-indicator, .theme-toggle-icon, .hero-portrait-frame { animation: none; transition: none; }
    ::view-transition-group(root), ::view-transition-old(root), ::view-transition-new(root) { animation:
none; }
}
```

resources/js/app.jsx

Bootstraps React and resolves Inertia page components.

```
import './bootstrap';
import { createInertiaApp } from '@inertiajs/react';
import { createRoot } from 'react-dom/client';

const pages = import.meta.glob('./Pages/**/*.jsx', { eager: true });

createInertiaApp({
    resolve: (name) => {
        const page = pages[`./Pages/${name}.jsx`];
        if (!page) throw new Error(`Unknown Inertia page: ${name}`);
        return page;
    },
    setup({ el, App, props }) {
        createRoot(el).render(<App {...props} />);
    },
});
```

resources/data/portfolio.php

Stores static portfolio details such as education, skills, navigation, and contact links.

```
<?php
return [
    'nav' => [
        ['id' => 'home',      'label' => 'Home',      'footer' => 'Home'],
        ['id' => 'about',     'label' => 'About',     'footer' => 'About me'],
        ['id' => 'education', 'label' => 'Education', 'footer' => 'Education'],
        ['id' => 'skills',    'label' => 'Skills',    'footer' => 'Skills'],
        ['id' => 'projects',  'label' => 'Projects',  'footer' => 'Projects'],
        ['id' => 'contact',   'label' => 'Contact',   'footer' => 'Contact'],
    ],

    'education' => [
        ['years' => '2024 - 2028', 'title' => 'BS Information Technology', 'text' => 'Urdaneta City
University, Pangasinan'],
        ['years' => '2024',        'title' => 'Certification',             'text' => 'NCII Computer
System Servicing'],
        ['years' => '2023 - 2024', 'title' => 'Layout Artist',             'text' => 'A good layout
turns simple ideas into great designs.'],
        ['years' => '2022 - 2024', 'title' => 'Senior high school',        'text' => 'San Pedro National
High School, Computer System Servicing.'],
    ],

    'skills' => [
        '3D' => [
            ['Blender 3D', 'Modeling'],
            ['Unity 3D', 'Game Mechanics'],
            ['SketchUp Pro', 'Spatial Interiors'],
            ['C# Scripting', 'OOP'],
        ],
        'Creative & UI/UX' => [
            ['Figma', 'Auto-Layout / Prototyping'],
            ['Photoshop', 'Concept / Grading'],
            ['Illustrator', 'Vectors / Identity'],
            ['Design Systems', 'Token Architecture'],
        ],
        'Web & Mobile' => [
            ['Flutter & Dart', 'Cross-Platform'],
            ['React / Next.js', 'App Router / SSR'],
            ['HTML & CSS', 'Markup & Styling'],
            ['Tailwind CSS', 'Utility-First'],
        ],
        'Vibe & AI Stack' => [
            ['Prompt Writing', 'Effective Prompts'],
            ['Debugging', 'Read & Trace'],
            ['Git & GitHub', 'Version Control'],
        ],
        'Soft Skills' => [
            ['Communication', 'Clear updates'],
            ['Time Management', 'Meets deadlines'],
            ['Teamwork', 'Group projects'],
            ['Problem Solving', 'Step by step'],
        ],
    ],

    'contact' => [
        'email'    => 'markdjvillar0923@gmail.com',
        'facebook' => ['label' => 'Deeh Jay', 'url' => 'https://www.facebook.com/markdj.villar.7'],
        'github'   => ['label' => 'github.com/Dj-0923', 'url' => 'https://github.com/Dj-0923'],
        'figma'    => 'https://www.figma.com/design/bcU1lWdrzcK6gC0SIAYZ63/Lendy-Prototype?node-id=0-1',
    ],
];
```

resources/js/Pages/Portfolio.jsx

Renders the public portfolio and connects the dynamic database-backed projects section.

```
import { Head } from '@inertiajs/react';
import { useState } from 'react';
import ContactForm from '../Components/ContactForm';
import Footer from '../Components/Footer';
import Navbar from '../Components/Navbar';
import HeroPortrait from '../Components/HeroPortrait';
import ProjectsSection from '../Components/ProjectsSection';
import SectionTitle from '../Components/SectionTitle';

const aboutCards = [
    ['Who I Am', 'Mark-dj Villar, a third-year BSIT student. I love turning ideas into working projects
and learning something new with every build.'],
    ['What I Enjoy', 'I enjoy exploring new technologies, solving complex problems, and creating
innovative solutions.'],
    ['What I Study', 'BS Information Technology at Urdaneta City University, with hands-on work in web
development, 3D modeling, vector design, and UI prototyping.'],
    ['Beyond the Screen', 'When I’m not building, I’m exploring new tools, watching tutorials, and
playing online games. Always curious, always learning.'],
];

export default function Portfolio({ projects, portfolio }) {
    const [theme, setTheme] = useState(() => document.documentElement.dataset.theme === 'light' ?
'light' : 'dark');

    return (
        <>
            <Head title="Mark-dj Villar | Portfolio">
                <meta name="description" content="Portfolio of Mark-dj Villar, a third-year BSIT student
building websites, apps, 3D scenes, and vector art." />
            </Head>
            <a href="#main" className="sr-only focus:not-sr-only focus:fixed focus:left-3 focus:top-3
focus:z-50 focus:rounded-md focus:bg-coral focus:px-4 focus:py-2 focus:text-deep">Skip to content</a>
            <Navbar theme={theme} onThemeChange={setTheme} />
            <main id="main">
                <section id="home" aria-labelledby="hero-title" className="relative overflow-x-clip">
                    <div className="mx-auto grid max-w-7xl items-center gap-12 px-6 py-14 lg:
grid-cols-[1.25fr_1fr] lg:gap-16 lg:px-16 lg:py-20">
                        <div>
                            <p className="mb-4 font-sans text-sm font-bold uppercase tracking-widest
text-peach">Mark-dj Villar • Y3 BSIT • Web &amp;amp; Creative Developer</p>
                            <h1 id="hero-title" className="text-5xl font-extrabold leading-[1.05]
tracking-tighter sm:text-6xl lg:text-7xl">
                                Coding with <span className="italic text-coral">Rhythm</span>, Building
with Intuition &amp;amp; Precision.
                            </h1>
                            <p className="mt-8 max-w-xl text-lg leading-relaxed text-blush">
                                I&amp;apos;m a third-year BSIT student who turns ideas into working websites,
 apps, and 3D scenes. I care about clean code, clear design, and finishing what I start.
                            </p>
                            <div className="mt-10 flex flex-wrap gap-4">
                                <a href="#projects" className="inline-flex items-center gap-2 rounded-lg
bg-coral px-8 py-3.5 font-bold text-deep hover:opacity-90">
                                    Explore Projects <span aria-hidden="true">→</span>
                                </a>
                                <a href="#about" className="inline-flex items-center gap-2 rounded-lg
bg-panel px-8 py-3.5 font-bold hover:bg-white/15">
                                    About Me <span aria-hidden="true">→</span>
                                </a>
                            </div>
                        </div>
                        <HeroPortrait theme={theme} />
                    </div>
                </section>

                <section id="about" aria-labelledby="about-title" className="scroll-mt-20 px-6 py-20 lg:
px-16">
                    <div className="mx-auto max-w-7xl">
                        <SectionTitle id="about-title" title="About Me" subtitle="A third-year BSIT
student who learns by building, one small project at a time." subtitleClass="max-w-2xl" />
                        <div className="mt-10 grid gap-6 lg:grid-cols-[0.9fr_1.1fr]">
                            <article className="rounded-2xl border border-white/10 bg-panel p-6 lg:p-8">
                                <img src="/images/profile.png" alt="Portrait of Mark-dj Villar"
width="80" height="80" className="h-20 w-20 rounded-full border-2 border-coral bg-white object-cover" />
                                <h3 className="mt-6 text-2xl font-extrabold">Hi, I’m Mark-dj</h3>
                                <p className="mt-4 leading-relaxed text-blush">I’m a third-year BSIT
student who loves turning ideas into working projects. I enjoy 3D modeling, game design, and clean UI,
and I’m always learning something new with every build. Fast to start, careful to finish, and curious
about everything.</p>
                                <blockquote className="mt-6 rounded-xl border border-white/10 bg-ink
p-4">
                                    <p className="font-sans text-xs uppercase tracking-widest
text-coral">My goal</p>
                                    <p className="mt-2 text-blush">“Small edits today, bigger builds
tomorrow. I keep learning, one function and one fix at a time, until the code I read becomes the code I
can write.”</p>
                                </blockquote>
                            </article>
                            <div className="grid gap-4 sm:grid-cols-2">
                                {aboutCards.map(([heading, body]) => (
```

```
                                    <article key={heading} className="rounded-2xl border border-white/10
bg-panel p-6">
                                        <h3 className="text-lg font-extrabold">{heading}</h3>
                                        <p className="mt-3 text-sm leading-relaxed text-blush">{body}</p>
                                    </article>
                                ))}
                            </div>
                        </div>
                    </div>
                </section>

                <section id="education" aria-labelledby="education-title" className="scroll-mt-20 px-6
py-20 lg:px-16">
                    <div className="mx-auto max-w-7xl">
                        <SectionTitle id="education-title" title="Education" />
                        <ol className="mt-10 max-w-2xl space-y-10 border-l-2 border-coral pl-8">
                            {portfolio.education.map((item) => (
                                <li key={`${item.years}-${item.title}`} className="relative">
                                    <span aria-hidden="true" className="absolute -left-10 top-1.5 h-3.5
w-3.5 rounded-full bg-coral" />
                                    <p className="font-sans text-xs uppercase tracking-widest
text-peach">{item.years}</p>
                                    <h3 className="mt-1 text-xl font-extrabold">{item.title}</h3>
                                    <p className="mt-1 text-white">{item.text}</p>
                                </li>
                            ))}
                        </ol>
                    </div>
                </section>

                <section id="skills" aria-labelledby="skills-title" className="scroll-mt-20 px-6 py-20
lg:px-16">
                    <div className="mx-auto max-w-7xl">
                        <SectionTitle id="skills-title" title="Skills" subtitle="Categorized
capabilities spanning codebases, 3D engines, and graphic suites." subtitleClass="max-w-xl" />
                        <div className="mt-10 grid gap-5 sm:grid-cols-2 lg:grid-cols-3">
                            {Object.entries(portfolio.skills).map(([group, items]) => (
                                <article key={group} className="rounded-2xl border border-white/10
bg-panel p-6">
                                    <h3 className="text-xl font-extrabold">{group}</h3>
                                    <ul className="mt-5 space-y-2">
                                        {items.map(([name, note]) => (
                                            <li key={name} className="flex items-center justify-between
gap-3 rounded-lg border border-white/10 bg-ink px-3 py-2.5 text-sm">
                                                <span>{name}</span><span className="text-right font-sans
text-xs text-coral">{note}</span>
                                            </li>
                                        ))}
                                    </ul>
                                </article>
                            ))}
                        </div>
                    </div>
                </section>

                <ProjectsSection projects={projects} />

                <section id="contact" aria-labelledby="contact-title" className="scroll-mt-20 px-6 py-20
lg:px-16">
                    <div className="mx-auto max-w-7xl">
                        <SectionTitle id="contact-title" title="Let’s build something epic together."
subtitle="Whether you have an internship opening, an exciting freelance collaboration, or want to
explore vibe coding workflows, reach out!" subtitleClass="max-w-2xl" />
                        <div className="mt-10 grid gap-8 lg:grid-cols-[0.8fr_1.2fr]">
                            <div className="space-y-4">
                                <ul className="space-y-3">
                                    <li className="rounded-2xl border border-white/10 bg-panel p-4">
                                        <p className="font-sans text-xs uppercase tracking-widest
text-peach">Email</p>
                                        <a href={`mailto:${portfolio.contact.email}`} className="mt-1
block break-all font-bold no-underline">{portfolio.contact.email}</a>
                                    </li>
                                    <li className="rounded-2xl border border-white/10 bg-panel p-4">
                                        <p className="font-sans text-xs uppercase tracking-widest
text-peach">Social</p>
                                        <a href={portfolio.contact.facebook.url} target="_blank"
rel="noopener noreferrer" className="mt-1 block font-bold
no-underline">{portfolio.contact.facebook.label}</a>
                                    </li>
                                    <li className="rounded-2xl border border-white/10 bg-panel p-4">
                                        <p className="font-sans text-xs uppercase tracking-widest
text-peach">GitHub</p>
                                        <a href={portfolio.contact.github.url} target="_blank"
rel="noopener noreferrer" className="mt-1 block break-all font-bold
no-underline">{portfolio.contact.github.label}</a>
                                    </li>
                                </ul>
                            </div>
                            <ContactForm />
                        </div>
                    </div>
                </section>
            </main>
            <Footer portfolio={portfolio} />
        </>
```

```
    );
}
```

resources/js/Pages/Projects/Index.jsx

Displays project records and edit/delete actions.

```
import { Head, Link } from '@inertiajs/react';
import AdminLayout from '../../Components/AdminLayout';
import DeleteProjectButton from '../../Components/DeleteProjectButton';

export default function Index({ projects }) {
    return (
        <AdminLayout
            title="Projects"
            actions={<Link href="/projects/create" className="rounded-lg bg-coral px-5 py-2.5 text-sm
font-bold text-deep hover:opacity-90">+ New project</Link>}
        >
            <Head title="Manage projects" />
            <div className="overflow-x-auto rounded-2xl border border-white/10 bg-panel">
                <table className="w-full min-w-[720px] text-left text-sm">
                    <thead className="text-xs uppercase tracking-widest text-peach">
                        <tr>
                            <th className="p-4">Image</th>
                            <th className="p-4">Title</th>
                            <th className="p-4">Category</th>
                            <th className="p-4">Tools</th>
                            <th className="p-4">Completed</th>
                            <th className="p-4 text-right">Actions</th>
                        </tr>
                    </thead>
                    <tbody>
                        {projects.length === 0 && <tr><td colSpan="6" className="p-8 text-center
text-mute">No projects yet. Click “New project” to add one.</td></tr>}
                        {projects.map((project) => (
                            <tr key={project.id} className="border-t border-white/10">
                                <td className="p-4"><img src={project.image_url} alt="" className="h-12
w-20 rounded-md object-cover" /></td>
                                <td className="p-4 font-bold">{project.title}</td>
                                <td className="p-4">{project.category.name}</td>
                                <td className="p-4 text-blush">{project.tools}</td>
                                <td className="p-4">{project.date_completed}</td>
                                <td className="p-4">
                                    <div className="flex justify-end gap-2">
                                        <Link href={`/projects/${project.id}/edit`}
className="rounded-lg bg-white/10 px-3 py-1.5 text-sm font-bold hover:bg-white/20">   Edit</Link>
                                        <DeleteProjectButton project={project} />
                                    </div>
                                </td>
                            </tr>
                        ))}
                    </tbody>
                </table>
            </div>
        </AdminLayout>
    );
}
```

resources/js/Pages/Projects/Create.jsx

Renders the reusable form in create mode.

```
import { Head } from '@inertiajs/react';
import AdminLayout from '../../Components/AdminLayout';
import ProjectForm from '../../Components/ProjectForm';

export default function Create({ categories }) {
    return (
        <AdminLayout title="New project">
            <Head title="New project" />
            <ProjectForm categories={categories} />
        </AdminLayout>
    );
}
```

resources/js/Pages/Projects/Edit.jsx

Renders the reusable form populated with a selected project.

```
import { Head } from '@inertiajs/react';
import AdminLayout from '../../Components/AdminLayout';
import ProjectForm from '../../Components/ProjectForm';

export default function Edit({ project, categories }) {
    return (
        <AdminLayout title="Edit project">
            <Head title={`Edit ${project.title}`} />
            <ProjectForm project={project} categories={categories} />
        </AdminLayout>
    );
}
```

resources/js/Components/AdminLayout.jsx

Provides the shared project-management header, layout, and flash feedback.

```
import { Head, Link } from '@inertiajs/react';
import FlashMessage from './FlashMessage';

export default function AdminLayout({ title, actions, children }) {
    return (
        <div className="min-h-screen">
            <Head title={`${title} | Manage projects`} />
            <header className="sticky top-0 z-40 border-b border-white/10 bg-ink/90 backdrop-blur">
                <div className="mx-auto flex h-20 max-w-5xl items-center justify-between gap-4 px-6">
                    <Link href="/" className="flex items-center gap-3">
                        <img src="/images/logo.png" alt="" className="h-11 w-[54px] rounded-xl border
border-white/10 object-cover" />
                        <span className="leading-tight">
                            <span className="block text-base font-bold">Mark-dj Villar</span>
                            <span className="block text-[11px] font-medium uppercase tracking-wider
text-peach">Manage projects</span>
                        </span>
                    </Link>
                    <Link href="/" className="rounded-md border border-white/20 px-4 py-1.5 text-sm
font-bold hover:bg-white/10">
                        ← Back to portfolio
                    </Link>
                </div>
            </header>
            <main className="mx-auto max-w-5xl px-6 py-10">
                <div className="mb-8 flex flex-wrap items-center justify-between gap-3">
                    <h1 className="text-3xl font-extrabold tracking-tight">{title}</h1>
                    {actions}
                </div>
                <FlashMessage />
                {children}
            </main>
        </div>
    );
}
```

resources/js/Components/ConfirmDialog.jsx

Provides the accessible custom delete-confirmation dialog.

```
import { useEffect, useRef } from 'react';

export default function ConfirmDialog({ open, title, message, busy, onConfirm, onCancel }) {
    const cancelRef = useRef(null);

    useEffect(() => {
        if (!open) return undefined;
        cancelRef.current?.focus();
        const handleKeyDown = (event) => {
            if (event.key === 'Escape' && !busy) onCancel();
            if (event.key === 'Tab') {
                event.preventDefault();
                cancelRef.current?.focus();
            }
        };
        document.addEventListener('keydown', handleKeyDown);
        document.body.classList.add('overflow-hidden');
        return () => {
            document.removeEventListener('keydown', handleKeyDown);
            document.body.classList.remove('overflow-hidden');
        };
    }, [open, busy, onCancel]);

    if (!open) return null;

    return (
        <div
            className="fixed inset-0 z-[100] flex items-center justify-center bg-black/70 p-4
backdrop-blur-sm"
            role="dialog"
            aria-modal="true"
            aria-labelledby="delete-confirmation-title"
            aria-describedby="delete-confirmation-message"
            onMouseDown={(event) => {
                if (event.target === event.currentTarget && !busy) onCancel();
            }}
        >
            <div className="w-full max-w-md overflow-hidden rounded-2xl border border-white/10 bg-panel
shadow-2xl">
                <div className="flex items-start gap-4 p-6">
                    <div className="flex h-12 w-12 shrink-0 items-center justify-center rounded-full
bg-red-500/10 text-red-400">
                        <svg aria-hidden="true" focusable="false" className="h-6 w-6" viewBox="0 0 24
24" fill="none" stroke="currentColor" strokeWidth="1.8" strokeLinecap="round" strokeLinejoin="round">
                            <path d="M12 9v4m0 4h.01" />
                            <path d="M10.3 3.9 1.8 18.2A2 2 0 0 0 3.5 21h17a2 2 0 0 0 1.7-2.8L13.7 3.9a2
2 0 0 0-3.4 0Z" />
                        </svg>
                    </div>
                    <div className="min-w-0 flex-1 text-left">
                        <h2 id="delete-confirmation-title" className="text-xl font-extrabold">{title}</
h2>
                        <p id="delete-confirmation-message" className="mt-2 text-left text-sm
leading-relaxed text-blush">{message}</p>
                    </div>
                </div>
                <div className="flex justify-end gap-3 border-t border-white/10 bg-black/10 px-6 py-4">
                    <button ref={cancelRef} type="button" onClick={onCancel} disabled={busy}
className="rounded-lg border border-white/20 px-4 py-2 text-sm font-bold hover:bg-white/10 disabled:
opacity-60">
                        Cancel
                    </button>
                    <button type="button" onClick={onConfirm} disabled={busy} className="rounded-lg
bg-red-700 px-4 py-2 text-sm font-bold text-white hover:bg-red-600 disabled:opacity-60">
                        {busy ? 'Deleting…' : 'Delete project'}
                    </button>
                </div>
            </div>
        </div>
    );
}
```

resources/js/Components/ContactForm.jsx

Implements the portfolio contact form client-side checks.

```
import { useState } from 'react';
import FormField, { inputClass } from './FormField';

const emptyValues = { name: '', email: '', subject: '', message: '' };
const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

export default function ContactForm() {
    const [values, setValues] = useState(emptyValues);
    const [errors, setErrors] = useState({});
    const [status, setStatus] = useState('');
    const update = (key) => (event) => setValues((current) => ({ ...current, [key]: event.target.value
}));

    const submit = (event) => {
        event.preventDefault();
        setStatus('');
        const nextErrors = {};
        if (values.name.trim().length < 2) nextErrors.name = 'Enter your name (at least 2 characters).';
        if (!emailPattern.test(values.email.trim())) nextErrors.email = 'Enter a valid email, like
name@example.com.';
        if (!values.subject) nextErrors.subject = 'Choose what this is about.';
        if (values.message.trim().length < 10) nextErrors.message = 'Write a message of at least 10
characters.';
        setErrors(nextErrors);
        if (Object.keys(nextErrors).length) {
            document.getElementById(Object.keys(nextErrors)[0])?.focus();
            return;
        }
        setStatus(`Thanks, ${values.name.trim()}! Your message is ready (this demo does not send it).`);
        setValues(emptyValues);
    };

    return (
        <form id="contact-form" noValidate onSubmit={submit} className="rounded-2xl border border-white/
10 bg-panel p-6 sm:p-8">
            <FormField id="name" label="Your name" error={errors.name}>
                <input id="name" type="text" autoComplete="name" placeholder="e.g. Dj Villar"
value={values.name} onChange={update('name')} aria-invalid={!!errors.name} aria-describedby="name-error"
className={inputClass(errors.name)} />
            </FormField>
            <FormField id="email" label="Your email" error={errors.email}>
                <input id="email" type="email" autoComplete="email" placeholder="dj@company.com"
value={values.email} onChange={update('email')} aria-invalid={!!errors.email}
aria-describedby="email-error" className={inputClass(errors.email)} />
            </FormField>
            <FormField id="subject" label="Subject / opportunity type" error={errors.subject}>
                <select id="subject" value={values.subject} onChange={update('subject')}
aria-invalid={!!errors.subject} aria-describedby="subject-error" className={inputClass(errors.subject)}>
                    <option value="">Choose one</option>
                    <option>Freelance Project</option>
                    <option>Collaboration</option>
                    <option>Just saying hi</option>
                </select>
            </FormField>
            <FormField id="message" label="Your message" error={errors.message}>
                <textarea id="message" rows="5" placeholder="Tell me about your vision, company, or
project requirements..." value={values.message} onChange={update('message')}
aria-invalid={!!errors.message} aria-describedby="message-error" className={inputClass(errors.message)} /
>
            </FormField>
            <button type="submit" className="w-full rounded-lg bg-linear-to-r from-coral to-peach px-6
py-3.5 font-bold text-deep hover:opacity-90">Send Message</button>
            <p role="status" className="mt-4 font-bold text-peach">{status}</p>
        </form>
    );
}
```

resources/js/Components/DeleteProjectButton.jsx

Connects the confirmation dialog to Inertia delete requests.

```
import { router } from '@inertiajs/react';
import { useState } from 'react';
import ConfirmDialog from './ConfirmDialog';

export default function DeleteProjectButton({ project }) {
    const [open, setOpen] = useState(false);
    const [busy, setBusy] = useState(false);

    const destroy = () => {
        router.delete(`/projects/${project.id}`, {
            preserveScroll: true,
            onStart: () => setBusy(true),
            onFinish: () => {
                setBusy(false);
                setOpen(false);
            },
        });
    };

    return (
        <>
            <button type="button" onClick={() => setOpen(true)} className="rounded-lg bg-red-700 px-3
py-1.5 text-sm font-bold text-white hover:bg-red-600">
                  Delete
            </button>
            <ConfirmDialog
                open={open}
                busy={busy}
                title="Confirmation"
                message={`Are you sure you want to delete “${project.title}”? This action cannot be
undone.`}
                onConfirm={destroy}
                onCancel={() => setOpen(false)}
            />
        </>
    );
}
```

resources/js/Components/FlashMessage.jsx

Displays success feedback shared by Laravel middleware.

```
import { usePage } from '@inertiajs/react';

export default function FlashMessage() {
    const message = usePage().props.flash?.success;
    if (!message) return null;

    return (
        <p role="status" className="mb-6 rounded-xl border border-green-500/40 bg-green-500/10 px-4 py-3
text-sm font-bold text-green-500">
            {message}
        </p>
    );
}
```

resources/js/Components/Footer.jsx

Renders the shared portfolio footer and navigation.

```
export default function Footer({ portfolio }) {
    return (
        <footer className="border-t border-white/10 px-6 py-12 lg:px-16">
            <div className="mx-auto grid max-w-7xl gap-10 md:grid-cols-[1.5fr_1fr_1fr]">
                <div>
                    <p className="text-xl font-extrabold">Mark-dj Villar</p>
                    <p className="mt-3 max-w-sm text-sm leading-relaxed text-blush">3rd year IT student
and creative vibe coder engineering high-impact web interactions, experimental visual systems, and
performant digital experiences.</p>
                </div>
                <nav aria-label="Footer">
                    <p className="font-sans text-xs uppercase tracking-widest text-peach">Quick
navigation</p>
                    <ul className="mt-3 space-y-2 text-sm">
                        {portfolio.nav.map((item) => <li key={item.id}><a href={`#${item.id}`}
className="text-blush hover:text-white">{item.footer}</a></li>)}
                    </ul>
                </nav>
                <div>
                    <p className="font-sans text-xs uppercase tracking-widest text-peach">Connect</p>
                    <ul className="mt-3 space-y-2 text-sm">
                        <li><a href={`mailto:${portfolio.contact.email}`} className="text-blush hover:
text-white">Email</a></li>
                        <li><a href={portfolio.contact.github.url} target="_blank" rel="noopener
noreferrer" className="text-blush hover:text-white">GitHub</a></li>
                    </ul>
                </div>
            </div>
            <div className="mx-auto mt-10 flex max-w-7xl flex-col gap-2 border-t border-white/10 pt-6
text-sm text-mute sm:flex-row sm:justify-between">
                <p>&copy; <span>{new Date().getFullYear()}</span> Mark-dj Villar. Vibe Coder.</p>
                <p><a href="#main" className="underline underline-offset-4">Back to top</a></p>
            </div>
        </footer>
    );
}
```

resources/js/Components/FormField.jsx

Reuses labeled form fields and adjacent validation-error display.

```
export const inputClass = (error) =>
    `mt-2 w-full rounded-lg border bg-ink px-4 py-2.5 text-white placeholder:text-mute ${
        error ? 'border-red-400' : 'border-white/30'
    }`;

export default function FormField({ id, label, error, children, className = '' }) {
    return (
        <div className={className}>
            <label htmlFor={id} className="block font-sans text-xs font-medium uppercase tracking-widest
text-peach">
                {label}
            </label>
            {children}
            <p id={`${id}-error`} role={error ? 'alert' : undefined} className="mt-1 min-h-5 text-sm
font-bold text-red-400">
                {error}
            </p>
        </div>
    );
}
```

resources/js/Components/HeroPortrait.jsx

Displays and transitions the portrait for light/dark theme changes.

```
import { useEffect, useRef } from 'react';

const portraitImages = {
    dark: '/images/no_sunglasses.png',
    light: '/images/with_sunglasses.png',
};
const portraitVideos = {
    dark: '/images/darkmode_web.mp4',
    light: '/images/lightmode_web.mp4',
};

export default function HeroPortrait({ theme }) {
    const imageRef = useRef(null);
    const videoRef = useRef(null);
    const initialTheme = useRef(theme);

    useEffect(() => {
        const image = imageRef.current;
        const video = videoRef.current;
        if (!image || !video) return;
        const source = portraitImages[theme];

        const showImage = () => {
            image.src = source;
            image.hidden = false;
            video.pause();
            video.hidden = true;
        };

        if (initialTheme.current === theme) {
            showImage();
            return;
        }
        initialTheme.current = theme;
        video.src = portraitVideos[theme];
        video.hidden = false;
        image.hidden = true;
        video.onended = showImage;
        video.onerror = () => {
            console.warn(`Unable to play the ${theme} mode portrait transition.`);
            showImage();
        };
        video.load();
        video.play().catch((error) => {
            console.warn(`Unable to play the ${theme} mode portrait transition.`, error);
            showImage();
        });
        return () => {
            video.pause();
            video.onended = null;
            video.onerror = null;
        };
    }, [theme]);

    return (
        <div className="relative mx-auto w-full max-w-sm lg:max-w-none">
            <div aria-hidden="true" className="absolute -inset-6 rounded-[3rem] bg-coral/40 blur-3xl" />
            <div className="hero-portrait-frame relative rounded-[2.5rem] bg-panel p-3.5">
                <div className="relative aspect-338/455 w-full overflow-hidden rounded-[1.75rem]
bg-white">
                    <img ref={imageRef} src={portraitImages[theme]} alt="Portrait of Mark-dj Villar"
width="338" height="455" className="absolute inset-0 h-full w-full object-cover" />
                    <video ref={videoRef} aria-hidden="true" muted playsInline preload="none" hidden
className="absolute inset-0 h-full w-full object-cover" />
                </div>
            </div>
        </div>
    );
}
```

resources/js/Components/Navbar.jsx

Implements portfolio navigation, mobile menu, active section, and theme toggle.

```
import { useEffect, useRef, useState } from 'react';

const links = [
    { id: 'home', label: 'Home' },
    { id: 'about', label: 'About' },
    { id: 'education', label: 'Education' },
    { id: 'skills', label: 'Skills' },
    { id: 'projects', label: 'Projects' },
    { id: 'contact', label: 'Contact' },
];

export default function Navbar({ theme, onThemeChange }) {
    const [menuOpen, setMenuOpen] = useState(false);
    const [activeSection, setActiveSection] = useState('home');
    const navRef = useRef(null);
    const navListRef = useRef(null);
    const indicatorRef = useRef(null);

    useEffect(() => {
        const updateActiveSection = () => {
            let current = 'home';
            links.forEach(({ id }) => {
                const section = document.getElementById(id);
                if (section && section.getBoundingClientRect().top <= window.innerHeight * 0.35) current
= id;
            });
            setActiveSection(current);
        };
        window.addEventListener('scroll', updateActiveSection, { passive: true });
        window.addEventListener('resize', updateActiveSection);
        updateActiveSection();
        return () => {
            window.removeEventListener('scroll', updateActiveSection);
            window.removeEventListener('resize', updateActiveSection);
        };
    }, []);

    useEffect(() => {
        const activeLink = navRef.current?.querySelector(`[data-nav="${activeSection}"]`);
        const list = navListRef.current;
        const indicator = indicatorRef.current;
        if (!activeLink || !list || !indicator || !navRef.current?.offsetWidth) return;
        const listRect = list.getBoundingClientRect();
        const linkRect = activeLink.getBoundingClientRect();
        indicator.style.width = `${linkRect.width}px`;
        indicator.style.height = `${linkRect.height}px`;
        indicator.style.transform = `translate3d(${linkRect.left - listRect.left}px, ${linkRect.top -
listRect.top}px, 0)`;
        indicator.classList.add('is-visible');
    }, [activeSection, menuOpen]);

    const changeTheme = () => {
        const nextTheme = theme === 'light' ? 'dark' : 'light';
        const apply = () => {
            document.documentElement.dataset.theme = nextTheme;
            onThemeChange(nextTheme);
            try {
                localStorage.setItem('portfolio-theme', nextTheme);
            } catch (error) {
                console.warn('Unable to save the selected color theme.', error);
            }
        };
        const reduceMotion = window.matchMedia('(prefers-reduced-motion: reduce)').matches;
        if (!reduceMotion && typeof document.startViewTransition === 'function') {
            const rect = document.getElementById('theme-toggle')?.getBoundingClientRect();
            if (rect) {
                document.documentElement.style.setProperty('--theme-switch-x', `${rect.left + rect.width
/ 2}px`);
                document.documentElement.style.setProperty('--theme-switch-y', `${rect.top + rect.height
/ 2}px`);
            }
            document.startViewTransition(apply);
        } else {
            apply();
        }
    };

    return (
        <header className="sticky top-0 z-40 border-b border-white/10 bg-ink/90 backdrop-blur">
            <div className="relative mx-auto grid h-20 max-w-7xl grid-cols-[minmax(0,1fr)_auto]
items-center px-6 lg:grid-cols-[auto_minmax(0,1fr)_auto] lg:px-16">
                <a href="#home" onClick={() => setMenuOpen(false)} className="flex items-center gap-3">
                    <img src="/images/logo.png" alt="" className="h-11 w-[54px] rounded-xl border
border-white/10 object-cover" />
                    <span className="leading-tight">
                        <span className="block text-base font-bold">Mark-dj Villar</span>
                        <span className="block font-sans text-[11px] font-medium uppercase
tracking-wider text-peach">Y-3 BSIT</span>
                    </span>
```

```
                </a>
                <nav ref={navRef} id="site-nav" aria-label="Main" className={`${menuOpen ? 'block' :
'hidden'} absolute right-6 top-full w-64 max-w-[calc(100%-3rem)] rounded-b-xl border border-t-0
border-white/10 bg-ink px-3 py-3 shadow-2xl lg:static lg:col-start-2 lg:block lg:w-auto lg:max-w-none lg:
justify-self-center lg:border-none lg:bg-transparent lg:p-0 lg:shadow-none`}>
                    <ul ref={navListRef} className="relative flex flex-col gap-1 lg:flex-row lg:
rounded-xl lg:bg-panel lg:p-1">
                        <span ref={indicatorRef} aria-hidden="true" className="nav-indicator" />
                        {links.map(({ id, label }) => (
                            <li key={id}>
                                <a
                                    data-nav={id}
                                    href={`#${id}`}
                                    aria-current={activeSection === id ? 'true' : undefined}
                                    onClick={() => setMenuOpen(false)}
                                    className={`nav-link block rounded-lg px-4 py-2 text-sm text-mute
hover:text-white aria-[current=true]:font-bold aria-[current=true]:text-deep lg:text-base
${activeSection === id ? 'font-bold text-deep' : ''}`}
                                >
                                    {label}
                                </a>
                            </li>
                        ))}
                    </ul>
                </nav>
                <div className="col-start-2 flex items-center justify-self-end gap-3 md:col-start-3">
                    <a href="#contact" className="hidden rounded-md bg-coral px-4 py-1.5 text-sm
font-bold text-deep hover:opacity-90 lg:inline-block">Get in Touch</a>
                    <button id="menu-toggle" type="button" aria-label={menuOpen ? 'Close navigation
menu' : 'Open navigation menu'} aria-expanded={menuOpen} aria-controls="site-nav" onClick={() =>
setMenuOpen((open) => !open)} className="rounded-md border border-white/20 p-2 lg:hidden">
                        <svg aria-hidden="true" className="h-5 w-5" viewBox="0 0 24 24" fill="none"
stroke="currentColor" strokeWidth="2" strokeLinecap="round">
                            <path d="M4 6h16M4 12h16M4 18h16" />
                        </svg>
                    </button>
                    <img src="/images/profile.png" alt="" width="32" height="32" className="h-10 w-10
shrink-0 rounded-full border border-white/20 object-cover" />
                    <button id="theme-toggle" type="button" aria-label={`Switch to ${theme === 'light' ?
'dark' : 'light'} mode`} aria-pressed={theme === 'light'} onClick={changeTheme} className="rounded-md
border border-white/20 p-2 text-blush transition-colors hover:bg-white/10" title={`Switch to ${theme ===
'light' ? 'dark' : 'light'} mode`}>
                        <svg className="theme-toggle-icon theme-icon-moon h-5 w-5" aria-hidden="true"
viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLinecap="round"
strokeLinejoin="round">
                            <path d="M20.9 13A9 9 0 0 1 11 3.1 9 9 0 1 0 20.9 13Z" />
                        </svg>
                        <svg className="theme-toggle-icon theme-icon-sun h-5 w-5" aria-hidden="true"
viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLinecap="round"
strokeLinejoin="round">
                            <circle cx="12" cy="12" r="4" />
                            <path d="M12 2v2m0 16v2M4.93 4.93l1.42 1.42m11.3 11.3 1.42 1.42M2 12h2m16
0h2M4.93 19.07l1.42-1.42m11.3-11.3 1.42-1.42" />
                        </svg>
                    </button>
                </div>
            </div>
        </header>
    );
}
```

resources/js/Components/ProjectCard.jsx

Renders project details, links, edit, and delete actions.

```
import { Link } from '@inertiajs/react';
import DeleteProjectButton from './DeleteProjectButton';

export default function ProjectCard({ project }) {
    const tools = project.tools.split(',').map((tool) => tool.trim()).filter(Boolean);

    return (
        <article className="overflow-hidden rounded-2xl border border-white/10 bg-panel">
            <img src={project.image_url} alt={`Preview of ${project.title}`} width="640" height="360"
loading="lazy" className="aspect-video w-full object-cover" />
            <div className="p-5">
                <p className="font-sans text-xs font-bold uppercase tracking-widest
text-coral">{project.category.name}</p>
                <h3 className="mt-1 text-lg font-extrabold">{project.title}</h3>
                <p className="mt-2 text-sm leading-relaxed text-blush">{project.description}</p>
                <ul className="mt-4 flex flex-wrap gap-2" aria-label="Tools used">
                    {tools.map((tool) => <li key={tool} className="rounded-full border border-white/15
bg-ink px-3 py-1 font-sans text-xs text-peach">{tool}</li>)}
                </ul>
                {(project.project_url || project.repository_url) && (
                    <p className="mt-4 flex gap-4 text-sm font-bold">
                        {project.project_url && <a href={project.project_url} target="_blank"
rel="noopener noreferrer" className="text-coral underline underline-offset-4">Live  </a>}
                        {project.repository_url && <a href={project.repository_url} target="_blank"
rel="noopener noreferrer" className="text-coral underline underline-offset-4">Code  </a>}
                    </p>
                )}
                <div className="mt-5 flex gap-2 border-t border-white/10 pt-4">
                    <Link href={`/projects/${project.id}/edit`} className="rounded-lg bg-white/10 px-3
py-1.5 text-sm font-bold hover:bg-white/20">   Edit</Link>
                    <DeleteProjectButton project={project} />
                </div>
            </div>
        </article>
    );
}
```

resources/js/Components/ProjectForm.jsx

Shares create/edit fields, file upload, category selection, and Inertia validation state.

```
import { Link, useForm } from '@inertiajs/react';
import { useState } from 'react';
import FormField, { inputClass } from './FormField';

const NEW_CATEGORY = '__new';
const now = new Date();
const today = `${now.getFullYear()}-${String(now.getMonth() + 1).padStart(2,
'0')}-${String(now.getDate()).padStart(2, '0')}`;

export default function ProjectForm({ project = null, categories }) {
    const isEdit = project !== null;
    const [imageName, setImageName] = useState('');
    const form = useForm({
        ...(isEdit ? { _method: 'put' } : {}),
        title: project?.title ?? '',
        category_id: project ? String(project.category_id) : categories.length ? '' : NEW_CATEGORY,
        new_category: '',
        description: project?.description ?? '',
        tools: project?.tools ?? '',
        project_url: project?.project_url ?? '',
        repository_url: project?.repository_url ?? '',
        date_completed: project?.date_completed ?? '',
        image: null,
    });
    const { data, setData, errors, processing } = form;
    const isNewCategory = data.category_id === NEW_CATEGORY;

    const submit = (event) => {
        event.preventDefault();
        form.transform((values) => ({
            ...values,
            category_id: values.category_id === NEW_CATEGORY ? '' : values.category_id,
            new_category: values.category_id === NEW_CATEGORY ? values.new_category : '',
        }));
        form.post(isEdit ? `/projects/${project.id}` : '/projects', {
            forceFormData: true,
            preserveScroll: true,
            onFinish: () => form.transform((values) => values),
        });
    };

    return (
        <form onSubmit={submit} encType="multipart/form-data" noValidate className="grid gap-x-6 gap-y-4
rounded-2xl border border-white/10 bg-panel p-6 sm:grid-cols-2 sm:p-8">
            <FormField id="title" label="Title" error={errors.title} className="sm:col-span-2">
                <input id="title" type="text" maxLength="60" required value={data.title}
onChange={(event) => setData('title', event.target.value)} placeholder="e.g. Figma - Lendy Prototype"
aria-describedby="title-error" className={inputClass(errors.title)} />
            </FormField>

            <FormField id="category_id" label="Category" error={errors.category_id}>
                <select id="category_id" value={data.category_id} onChange={(event) =>
setData('category_id', event.target.value)} aria-describedby="category_id-error"
className={inputClass(errors.category_id)}>
                    <option value="">Choose one</option>
                    {categories.map((category) => <option key={category.id}
value={category.id}>{category.name}</option>)}
                    <option value={NEW_CATEGORY}>+ Add new category…</option>
                </select>
            </FormField>

            <FormField id="date_completed" label="Date completed" error={errors.date_completed}>
                <div className="relative">
                    <input id="date_completed" type="date" max={today} required
value={data.date_completed} onChange={(event) => setData('date_completed', event.target.value)}
aria-describedby="date_completed-error" className={`${inputClass(errors.date_completed)}
date-completed-input pr-11`} />
                    <svg aria-hidden="true" focusable="false" className="pointer-events-none absolute
right-3 top-1/2 h-4 w-4 -translate-y-1/2 text-peach" viewBox="0 0 24 24" fill="none"
stroke="currentColor" strokeWidth="1.8" strokeLinecap="round" strokeLinejoin="round">
                        <rect x="3" y="5" width="18" height="16" rx="2" />
                        <path d="M16 3v4M8 3v4M3 11h18" />
                    </svg>
                </div>
            </FormField>

            {isNewCategory && (
                <FormField id="new_category" label="New category name" error={errors.new_category}
className="sm:col-span-2">
                    <input id="new_category" type="text" maxLength="30" required
value={data.new_category} onChange={(event) => setData('new_category', event.target.value)}
placeholder="e.g. Web" aria-describedby="new_category-error" className={inputClass(errors.new_category)}
/>
                </FormField>
            )}

            <FormField id="description" label="Description" error={errors.description} className="sm:
col-span-2">
                <textarea id="description" rows="4" minLength="20" maxLength="300" required
```

```
value={data.description} onChange={(event) => setData('description', event.target.value)}
aria-describedby="description-error" className={inputClass(errors.description)} />
            </FormField>

            <FormField id="tools" label="Tech stack / tools" error={errors.tools} className="sm:
col-span-2">
                <input id="tools" type="text" maxLength="80" required value={data.tools}
onChange={(event) => setData('tools', event.target.value)} placeholder="Flutter, Dart, Figma"
aria-describedby="tools-error" className={inputClass(errors.tools)} />
            </FormField>

            <FormField id="project_url" label="Project URL (optional)" error={errors.project_url}>
                <input id="project_url" type="url" maxLength="255" value={data.project_url}
onChange={(event) => setData('project_url', event.target.value)} placeholder="https://..."
aria-describedby="project_url-error" className={inputClass(errors.project_url)} />
            </FormField>

            <FormField id="repository_url" label="Repository URL (optional)"
error={errors.repository_url}>
                <input id="repository_url" type="url" maxLength="255" value={data.repository_url}
onChange={(event) => setData('repository_url', event.target.value)} placeholder="https://github.com/..."
aria-describedby="repository_url-error" className={inputClass(errors.repository_url)} />
            </FormField>

            <FormField id="image" label={isEdit ? 'Replace image (optional)' : 'Image'}
error={errors.image} className="sm:col-span-2">
                <input
                    id="image"
                    type="file"
                    accept="image/jpeg,image/png,image/webp"
                    required={!isEdit}
                    onChange={(event) => {
                        const file = event.target.files?.[0] ?? null;
                        setData('image', file);
                        setImageName(file?.name ?? '');
                    }}
                    aria-describedby="image-error"
                    className={`${inputClass(errors.image)} image-file-input`}
                />
                <p className="mt-1 text-sm text-mute">{imageName || 'JPG, PNG, or WEBP; maximum 2 MB.'}</
p>
                {isEdit && (
                    <div className="mt-3 flex items-center gap-3">
                        <img src={project.image_url} alt="" className="h-16 w-28 rounded-lg border
border-white/10 object-cover" />
                        <p className="text-sm text-mute">Current image. Choose a file only if you want
to replace it.</p>
                    </div>
                )}
            </FormField>

            <div className="mt-2 flex gap-3 sm:col-span-2">
                <button type="submit" disabled={processing} className="rounded-lg bg-coral px-6 py-2.5
font-bold text-deep hover:opacity-90 disabled:opacity-60">
                    {processing ? 'Saving…' : isEdit ? 'Update project' : 'Save project'}
                </button>
                <Link href="/projects" className="rounded-lg bg-white/10 px-6 py-2.5 font-bold hover:
bg-white/20">Cancel</Link>
            </div>
        </form>
    );
}
```

resources/js/Components/ProjectsSection.jsx

Renders category filters, project cards, empty state, and create link.

```
import { Link, usePage } from '@inertiajs/react';
import { useMemo, useState } from 'react';
import FlashMessage from './FlashMessage';
import ProjectCard from './ProjectCard';
import SectionTitle from './SectionTitle';

export default function ProjectsSection({ projects }) {
    const [filter, setFilter] = useState('all');
    const flash = usePage().props.flash?.success;
    const categories = useMemo(() => {
        const unique = new Map();
        projects.forEach((project) => unique.set(project.category.id, project.category));
        return [...unique.values()].sort((first, second) => first.name.localeCompare(second.name));
    }, [projects]);
    const visible = filter === 'all' ? projects : projects.filter((project) =>
String(project.category_id) === filter);

    return (
        <section id="projects" aria-labelledby="projects-title" className="scroll-mt-20 px-6 py-20 lg:
px-16">
            <div className="mx-auto max-w-7xl">
                <SectionTitle id="projects-title" title="Projects Accomplished" subtitle="Curated
creative suite and developer technologies." subtitleClass="max-w-xl" />
                {flash && <FlashMessage />}
                <div className="mt-8 flex flex-wrap items-center justify-between gap-4">
                    <div className="flex flex-wrap items-center gap-3" role="group" aria-label="Filter
projects by category">
                        {[{ id: 'all', name: 'All' }, ...categories.map((category) => ({ id:
String(category.id), name: category.name }))].map((category) => {
                            const active = filter === category.id;
                            return (
                                <button key={category.id} type="button" aria-pressed={active}
onClick={() => setFilter(category.id)} className={`rounded-full border px-4 py-2 text-sm font-bold hover:
text-white ${active ? 'border-coral bg-coral text-deep' : 'border-white/20 text-blush'}`}>
                                    {category.name}
                                </button>
                            );
                        })}
                    </div>
                    <Link href="/projects/create" className="rounded-lg bg-coral px-5 py-2.5 text-sm
font-bold text-deep hover:opacity-90">+ Add project</Link>
                </div>
                <p role="status" className="mt-3 text-sm text-mute">Showing {visible.length} of
{projects.length} projects</p>
                {projects.length === 0 ? (
                    <div className="mt-6 rounded-2xl border-2 border-dashed border-white/20 p-10
text-center">
                        <p className="text-4xl" aria-hidden="true"> </p>
                        <p className="mt-3 text-lg font-bold">No projects yet.</p>
                        <p className="mt-1 text-sm text-mute">Add a project to display it in your
portfolio.</p>
                    </div>
                ) : (
                    <div className="mt-6 grid gap-6 sm:grid-cols-2 lg:grid-cols-3">
                        {visible.map((project) => <ProjectCard key={project.id} project={project} />)}
                    </div>
                )}
            </div>
        </section>
    );
}
```

resources/js/Components/SectionTitle.jsx

Provides a reusable section heading and subtitle.

```
export default function SectionTitle({ id, title, subtitle, subtitleClass = '' }) {
    return (
        <>
            <h2 id={id} className="mt-4 max-w-3xl text-4xl font-extrabold leading-tight tracking-tight
sm:text-5xl">{title}</h2>
            {subtitle && <p className={`mt-4 text-lg text-blush ${subtitleClass}`}>{subtitle}</p>}
        </>
    );
}
```

vite.config.js

Configures Laravel, React JSX/refresh, and Tailwind CSS builds.

```
import { defineConfig } from 'vite';
import laravel from 'laravel-vite-plugin';
import react from '@vitejs/plugin-react';
import tailwindcss from '@tailwindcss/vite';

export default defineConfig({
    plugins: [
        react(),
        laravel({
            input: ['resources/css/app.css', 'resources/js/app.jsx'],
            refresh: true,
        }),
        tailwindcss(),
    ],
    server: {
        watch: {
            ignored: ['**/storage/framework/views/**'],
        },
    },
});
```

tests/Feature/ExampleTest.php

Verifies the public route returns the Portfolio Inertia page.

```
<?php

namespace Tests\Feature;

use Illuminate\Foundation\Testing\RefreshDatabase;
use Inertia\Testing\AssertableInertia as Assert;
use Tests\TestCase;

class ExampleTest extends TestCase
{
    use RefreshDatabase;

    /**
     * A basic test example.
     */
    public function test_the_application_returns_a_successful_response(): void
    {
        $this->get('/')
            ->assertOk()
            ->assertInertia(fn (Assert $page) => $page
                ->component('Portfolio')
                ->has('projects')
                ->has('portfolio.education')
                ->has('portfolio.skills'));
    }
}
```

tests/Feature/ProjectCrudTest.php

Covers create/read/update/delete, new categories, image storage, and success messages.

```
<?php

namespace Tests\Feature;

use App\Models\Category;
use App\Models\Project;
use Illuminate\Foundation\Testing\RefreshDatabase;
use Illuminate\Http\UploadedFile;
use Illuminate\Support\Facades\Storage;
use Inertia\Testing\AssertableInertia as Assert;
use Tests\TestCase;

class ProjectCrudTest extends TestCase
{
    use RefreshDatabase;

    public function test_a_project_can_be_created_updated_and_deleted(): void
    {
        Storage::fake('public');

        $image = UploadedFile::fake()->createWithContent(
            'project.png',
            base64_decode('iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAQAAAC1HAwCAAAAC0lEQVR42mP8/x8AAwMCAO+/
rWQAAAAASUVORK5CYII=')
        );

        $this->get(route('projects.create'))
            ->assertOk()
            ->assertInertia(fn (Assert $page) => $page->component('Projects/Create')->has('categories'));
        $this->post(route('projects.store'), [
            'title' => 'Portfolio project',
            'category_id' => '__new',
            'new_category' => 'Web',
            'description' => 'A sample project description long enough for validation.',
            'tools' => 'Laravel, PHP',
            'date_completed' => now()->toDateString(),
            'image' => $image,
        ])->assertRedirect(route('projects.index'))
            ->assertSessionHas('success', 'Project created successfully.');

        $project = Project::firstOrFail();
        $this->assertDatabaseHas('projects', [
            'id' => $project->id,
            'title' => 'Portfolio project',
        ]);
        $this->assertDatabaseHas('categories', ['name' => 'Web']);
        Storage::disk('public')->assertExists($project->image);

        $this->get(route('home'))
            ->assertOk()
            ->assertInertia(fn (Assert $page) => $page
                ->component('Portfolio')
                ->has('projects', 1)
                ->where('projects.0.title', 'Portfolio project'));
        $this->get(route('projects.index'))
            ->assertOk()
            ->assertInertia(fn (Assert $page) => $page
                ->component('Projects/Index')
                ->has('projects', 1));
        $this->get(route('projects.edit', $project))
            ->assertOk()
            ->assertInertia(fn (Assert $page) => $page
                ->component('Projects/Edit')
                ->where('project.title', 'Portfolio project')
                ->has('categories'));

        $category = Category::firstOrFail();
        $this->put(route('projects.update', $project), [
            'title' => 'Updated portfolio project',
            'category_id' => $category->id,
            'description' => 'An updated project description long enough for validation.',
            'tools' => 'Laravel, PHP, SQLite',
            'date_completed' => now()->toDateString(),
        ])->assertRedirect(route('projects.index'))
            ->assertSessionHas('success', 'Project updated successfully.');

        $this->assertDatabaseHas('projects', [
            'id' => $project->id,
            'title' => 'Updated portfolio project',
            'tools' => 'Laravel, PHP, SQLite',
        ]);

        $this->from(route('projects.index'))
            ->delete(route('projects.destroy', $project))
            ->assertRedirect(route('projects.index'))
            ->assertSessionHas('success', 'Project deleted successfully.');
        $this->assertDatabaseMissing('projects', ['id' => $project->id]);
        Storage::disk('public')->assertMissing($project->image);
    }
```

```
    public function test_duplicate_project_titles_are_rejected(): void
    {
        $category = Category::create(['name' => 'Web']);
        Project::create([
            'category_id' => $category->id,
            'title' => 'Existing project',
            'description' => 'An existing project description long enough.',
            'tools' => 'Laravel',
            'image' => 'projects/existing.png',
            'date_completed' => now()->toDateString(),
        ]);

        $this->from(route('projects.create'))->post(route('projects.store'), [
            'title' => 'Existing project',
            'category_id' => $category->id,
            'description' => 'A new project description long enough for validation.',
            'tools' => 'Laravel',
            'date_completed' => now()->toDateString(),
            'image' => UploadedFile::fake()->createWithContent(
                'project.png',
                base64_decode('iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAQAAAC1HAwCAAAAC0lEQVR42mP8/x8AAwMCAO+/
rWQAAAAASUVORK5CYII=')
            ),
        ])->assertRedirect(route('projects.create'))->assertSessionHasErrors('title');
    }
}
```

tests/Feature/ProjectValidationTest.php

Covers required fields, length, URL/date formats, image constraints, and update-safe uniqueness.

```
<?php

namespace Tests\Feature;

use App\Models\Category;
use App\Models\Project;
use Illuminate\Foundation\Testing\RefreshDatabase;
use Illuminate\Http\UploadedFile;
use Tests\TestCase;

class ProjectValidationTest extends TestCase
{
    use RefreshDatabase;

    public function test_required_project_fields_are_validated(): void
    {
        $this->from(route('projects.create'))
            ->post(route('projects.store'), [])
            ->assertRedirect(route('projects.create'))
            ->assertSessionHasErrors(['title', 'category_id', 'new_category', 'description', 'tools',
'date_completed', 'image']);
    }

    public function test_project_title_cannot_exceed_its_maximum_length(): void
    {
        $this->submitProject(['title' => str_repeat('x', 61)])
            ->assertSessionHasErrors('title');
    }

    public function test_updating_a_project_without_changing_its_unique_title_is_allowed(): void
    {
        $category = Category::create(['name' => 'Web']);
        $project = Project::create([
            'category_id' => $category->id,
            'title' => 'Existing project',
            'description' => 'An existing project description long enough.',
            'tools' => 'Laravel',
            'image' => 'projects/existing.png',
            'date_completed' => now()->toDateString(),
        ]);

        $this->put(route('projects.update', $project), [
            'title' => $project->title,
            'category_id' => $category->id,
            'description' => $project->description,
            'tools' => $project->tools,
            'date_completed' => $project->date_completed->format('Y-m-d'),
        ])->assertRedirect(route('projects.index'))
            ->assertSessionHasNoErrors();
    }

    public function test_project_url_must_be_a_valid_url(): void
    {
        $this->submitProject(['project_url' => 'not-a-url'])
            ->assertSessionHasErrors('project_url');
    }

    public function test_completion_date_cannot_be_in_the_future(): void
    {
        $this->submitProject(['date_completed' => now()->addDay()->toDateString()])
            ->assertSessionHasErrors('date_completed');
    }

    public function test_uploaded_project_image_must_be_an_image(): void
    {
        $this->submitProject([
            'image' => UploadedFile::fake()->create('notes.txt', 20, 'text/plain'),
        ])->assertSessionHasErrors('image');
    }

    public function test_uploaded_project_image_cannot_exceed_two_megabytes(): void
    {
        $png = base64_decode('iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAQAAAC1HAwCAAAAC0lEQVR42mP8/x8AAwMCAO+/
rWQAAAAASUVORK5CYII=');
        $this->submitProject([
            'image' => UploadedFile::fake()->createWithContent('large.png', $png.str_repeat('x', 2 *
1024 * 1024)),
        ])->assertSessionHasErrors('image');
    }

    private function submitProject(array $overrides = [])
    {
        $category = Category::firstOrCreate(['name' => 'Web']);
        $payload = array_merge([
            'title' => 'Validation project',
            'category_id' => $category->id,
            'description' => 'A valid project description long enough for validation.',
            'tools' => 'Laravel, PHP',
            'date_completed' => now()->toDateString(),
```

```
            'image' => UploadedFile::fake()->createWithContent(
                'project.png',
                base64_decode('iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAQAAAC1HAwCAAAAC0lEQVR42mP8/x8AAwMCAO+/
rWQAAAAASUVORK5CYII=')
            ),
        ], $overrides);

        return $this->from(route('projects.create'))->post(route('projects.store'), $payload);
    }
}
```

6. Design Decisions

Why use a separate resource controller?

The ProjectController keeps project-specific actions and validation out of the route file. Route::resource provides predictable index, create, store, edit, update, and destroy endpoints.

Why these validation rules and where are they placed?

Rules live in ProjectController so requests are checked on the server even when client-side controls are bypassed. The rules protect required fields, lengths, URLs, dates, uniqueness, category references, and image uploads.

Why organize React pages and reusable components?

Inertia Pages represent route-level screens while shared components hold repeated behavior such as ProjectForm, FormField, project cards, feedback, and confirmation. This keeps Create and Edit consistent while allowing Laravel to remain responsible for data and validation.

What happens without $fillable?

Eloquent mass-assignment protection rejects unapproved attributes when Project::create or update receives an array. The model explicitly lists fields that this form may assign.

How do validation errors reach the form?

Laravel redirects back with validation errors in the session. Inertia shares the errors on the next page response, and useForm exposes them beside each field without discarding the entered form state.

7. Validation Test Table

#	Test input	Expected result	Automated result
1	Submit all fields empty	Required-field errors appear.	Pass
2	Title contains 61 characters	Title maximum-length error appears.	Pass
3	Edit while keeping the same project title	Update is accepted; unique rule ignores this record.	Pass
4	Project URL is not a valid URL	URL format error appears.	Pass
5	Completion date is tomorrow	Future-date error appears.	Pass
6	Upload a text file as the image	Image type error appears.	Pass
7	Upload an image larger than 2 MB	Image size error appears.	Pass

Verification run: php artisan test ? 11 tests passed, 96 assertions. Validation-specific feature tests cover all seven cases above.

8. Reflection

Draft only: personalize these paragraphs to reflect your own experience before submitting.

Hardest part of the conversion

The hardest part was connecting the Laravel project data and validation flow to the React forms while preserving the original portfolio layout. I addressed this by keeping database operations in the resource controller and sharing form and display behavior through React components.

What I learned about validation

I learned that client-side validation improves usability, but server-side validation is still required because browser checks can be bypassed. Laravel can return field-specific errors through Inertia while the form keeps its entered values.

What I would change for a second dynamic section

If I added a second dynamic section, I would first define its data model and validation cases, then build a separate resource controller and reusable components. I would also add authorization before exposing management actions publicly.

Personalize the draft above so the submitted reflection is accurate and written in your own words.

9. Links and Final Submission

Item	Link / status
GitHub repository	[Paste the repository URL]
Video demonstration	[Paste a viewable Google Drive or unlisted YouTube URL]
Video requirements	Record your own voice; show CRUD, validation, and an uncut live field + validation-rule change.
Class section	[Fill in your section]
Folder-structure screenshot	[Add a screenshot from VS Code Explorer or your file browser]

The application and automated tests are prepared. The student must record the video, perform the live code change during the recording, personalize the reflection, add the remaining folder-structure screenshot, and supply shareable submission links.